Exercise 2F

1 Sketch the graphs of the following equations. For each graph, show the coordinates of the point(s) where the graph crosses the coordinate axes, and write down the coordinate of the turning point and the equation of the line of symmetry.

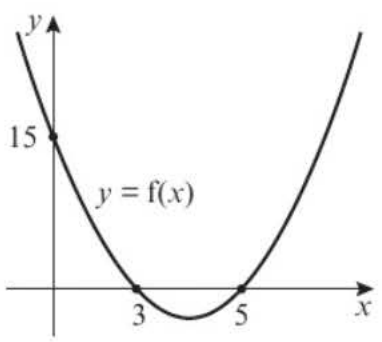
a $y = x^2 - 6x + 8$ **b** $y = x^2 + 2x - 15$ **c** $y = 25 - x^2$ **d** $y = x^2 + 3x + 2$

e $y = -x^2 + 6x + 7$ **f** $y = 2x^2 + 4x + 10$ **g** $y = 2x^2 + 7x - 15$ **h** $y = 6x^2 - 19x + 10$

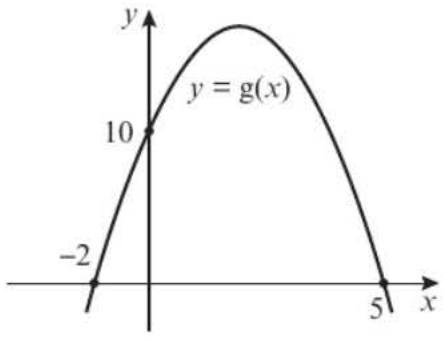
i $y = 4 - 7x - 2x^2$ **j** $y = 0.5x^2 + 0.2x + 0.02$

(P) **2** These sketches are graphs of quadratic functions of the form $ax^2 + bx + c$.
Find the values of a, b and c for each function.

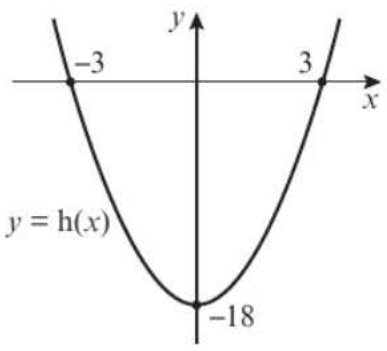
a



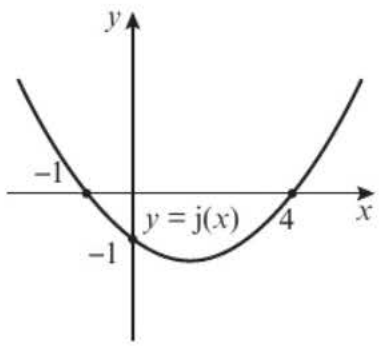
b



c



d



Problem-solving

Check your answers by substituting values into the function. In part **c** the graph passes through (0, –18), so h(0) should be –18.

(E/P) **3** The graph of $y = ax^2 + bx + c$ has a minimum at (5, –3) and passes through (4, 0).
Find the values of a, b and c. **(3 marks)**

2.5 The discriminant

If you square any real number, the result is greater than or equal to 0. This means that if y is negative, $\sqrt{y}$ cannot be a real number. Look at the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

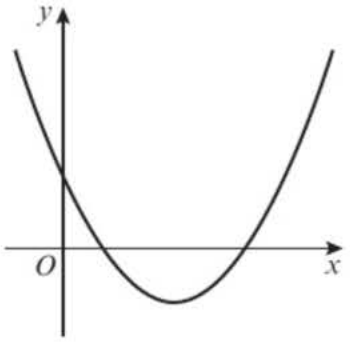
If the value under the square root sign is negative, x cannot be a real number and there are no real solutions. If the value under the square root is equal to 0, both solutions will be the same.

- **For the quadratic function $f(x) = ax^2 + bx + c$, the expression $b^2 - 4ac$ is called the discriminant. The value of the discriminant shows how many roots $f(x)$ has:**
  - **If $b^2 - 4ac > 0$ then $f(x)$ has two distinct real roots.**
  - **If $b^2 - 4ac = 0$ then $f(x)$ has one repeated root.**
  - **If $b^2 - 4ac < 0$ then $f(x)$ has no real roots.**

You can use the discriminant to check the shape of sketch graphs.

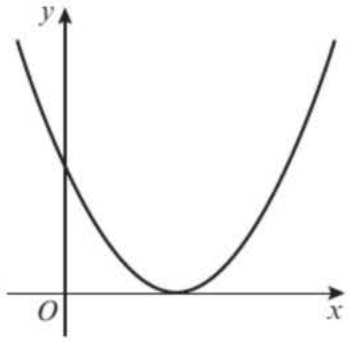
Below are some graphs of $y = f(x)$ where $f(x) = ax^2 + bx + c$.

$a > 0$



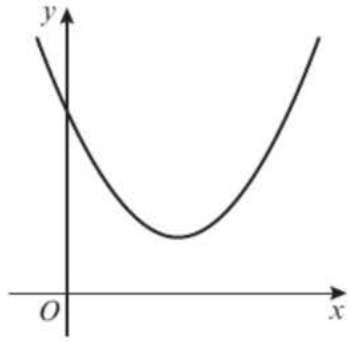
$b^2 - 4ac > 0$

Two distinct real roots



$b^2 - 4ac = 0$

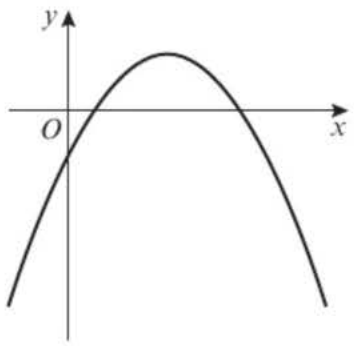
One repeated root

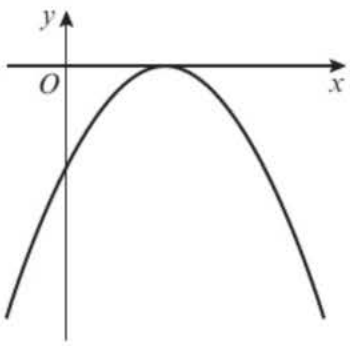


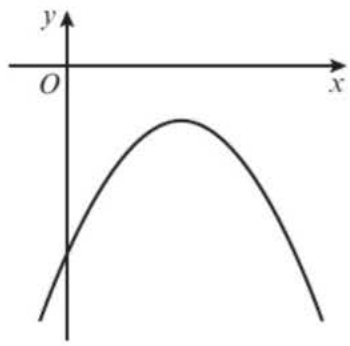
$b^2 - 4ac < 0$

No real roots

$a < 0$







Example 13

Find the values of k for which $f(x) = x^2 + kx + 9$ has equal roots.

$x^2 + kx + 9 = 0$
Here $a = 1$, $b = k$ and $c = 9$
For equal roots, $b^2 - 4ac = 0$
$k^2 - 4 \times 1 \times 9 = 0$
$k^2 - 36 = 0$
$k^2 = 36$
so $k = \pm 6$

Problem-solving

Use the condition given in the question to write a statement about the discriminant.

Substitute for a, b and c to get an equation with one unknown.

Solve to find the values of k.

Example 14

Find the range of values of k for which $x^2 + 4x + k = 0$ has two distinct real solutions.

$x^2 + 4x + k = 0$
Here $a = 1$, $b = 4$ and $c = k$.
For two real solutions, $b^2 - 4ac > 0$
$4^2 - 4 \times 1 \times k > 0$
$16 - 4k > 0$
$16 > 4k$
$4 > k$
So $k < 4$

This statement involves an inequality, so your answer will also be an inequality.

For any value of k less than 4, the equation will have 2 distinct real solutions.

Online Explore how the value of the discriminant changes with k using GeoGebra.

Exercise 2G

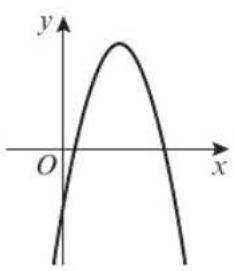
1 a Calculate the value of the discriminant for each of these five functions:

i $f(x) = x^2 + 8x + 3$ **ii** $g(x) = 2x^2 - 3x + 4$ **iii** $h(x) = -x^2 + 7x - 3$

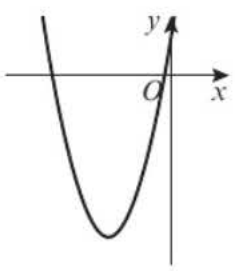
iv $j(x) = x^2 - 8x + 16$ **v** $k(x) = 2x - 3x^2 - 4$

b Using your answers to part **a**, match the same five functions to these sketch graphs.

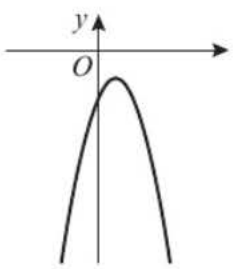
i



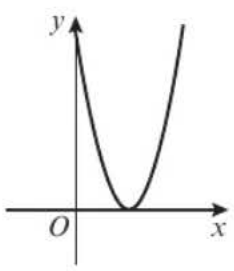
ii



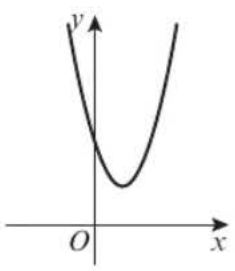
iii



iv



v



E/P **2** Find the values of k for which $x^2 + 6x + k = 0$ has two real solutions. **(2 marks)**

E/P **3** Find the value of t for which $2x^2 - 3x + t = 0$ has exactly one solution. **(2 marks)**

E/P **4** Given that the function $f(x) = sx^2 + 8x + s$ has equal roots, find the value of the positive constant s. **(2 marks)**

E/P **5** Find the range of values of k for which $3x^2 - 4x + k = 0$ has no real solutions. **(2 marks)**

E/P **6** The function $g(x) = x^2 + 3px + (14p - 3)$, where p is an integer, has two equal roots.

a Find the value of p. **(2 marks)**

b For this value of p, solve the equation $x^2 + 3px + (14p - 3) = 0$. **(2 marks)**

E/P **7** $h(x) = 2x^2 + (k + 4)x + k$, where k is a real constant.

a Find the discriminant of $h(x)$ in terms of k. **(3 marks)**

b Hence or otherwise, prove that $h(x)$ has two distinct real roots for all values of k. **(3 marks)**

Problem-solving

If a question part says 'hence or otherwise' it is usually easier to use your answer to the previous question part.

Challenge

a Prove that, if the values of a and c are given and non-zero, it is always possible to choose a value of b so that $f(x) = ax^2 + bx + c$ has distinct real roots.

b Is it always possible to choose a value of b so that $f(x)$ has equal roots? Explain your answer.

2.6 Modelling with quadratics

A **mathematical model** is a mathematical description of a real-life situation. Mathematical models use the language and tools of mathematics to represent and explore real-life patterns and relationships, and to predict what is going to happen next.

Models can be simple or complicated, and their results can be approximate or exact. Sometimes a model is only valid under certain circumstances, or for a limited range of inputs. You will learn more about how models involve simplifications and assumptions in Statistics and Mechanics.

Quadratic functions can be used to model and explore a range of practical contexts, including projectile motion.